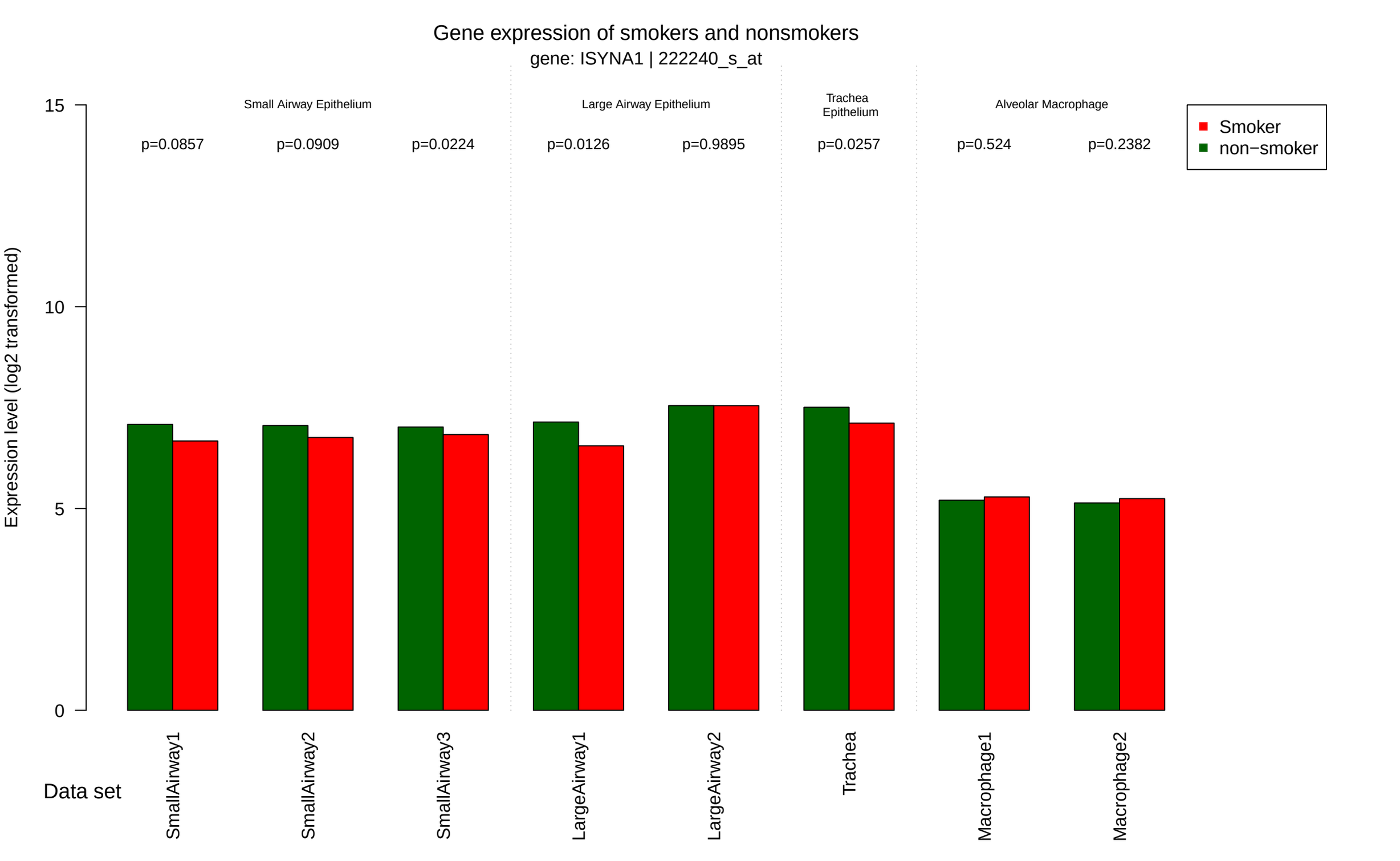
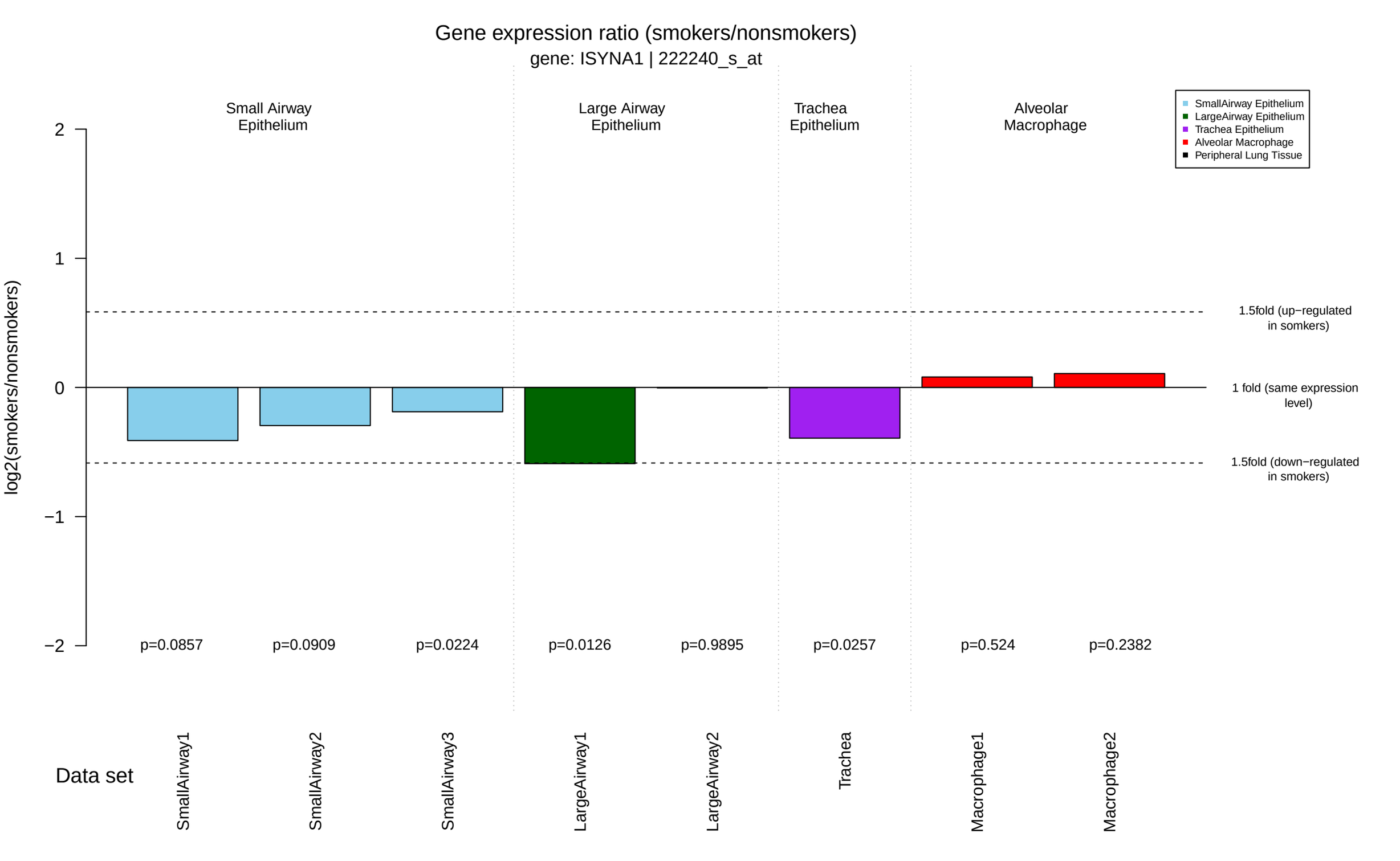
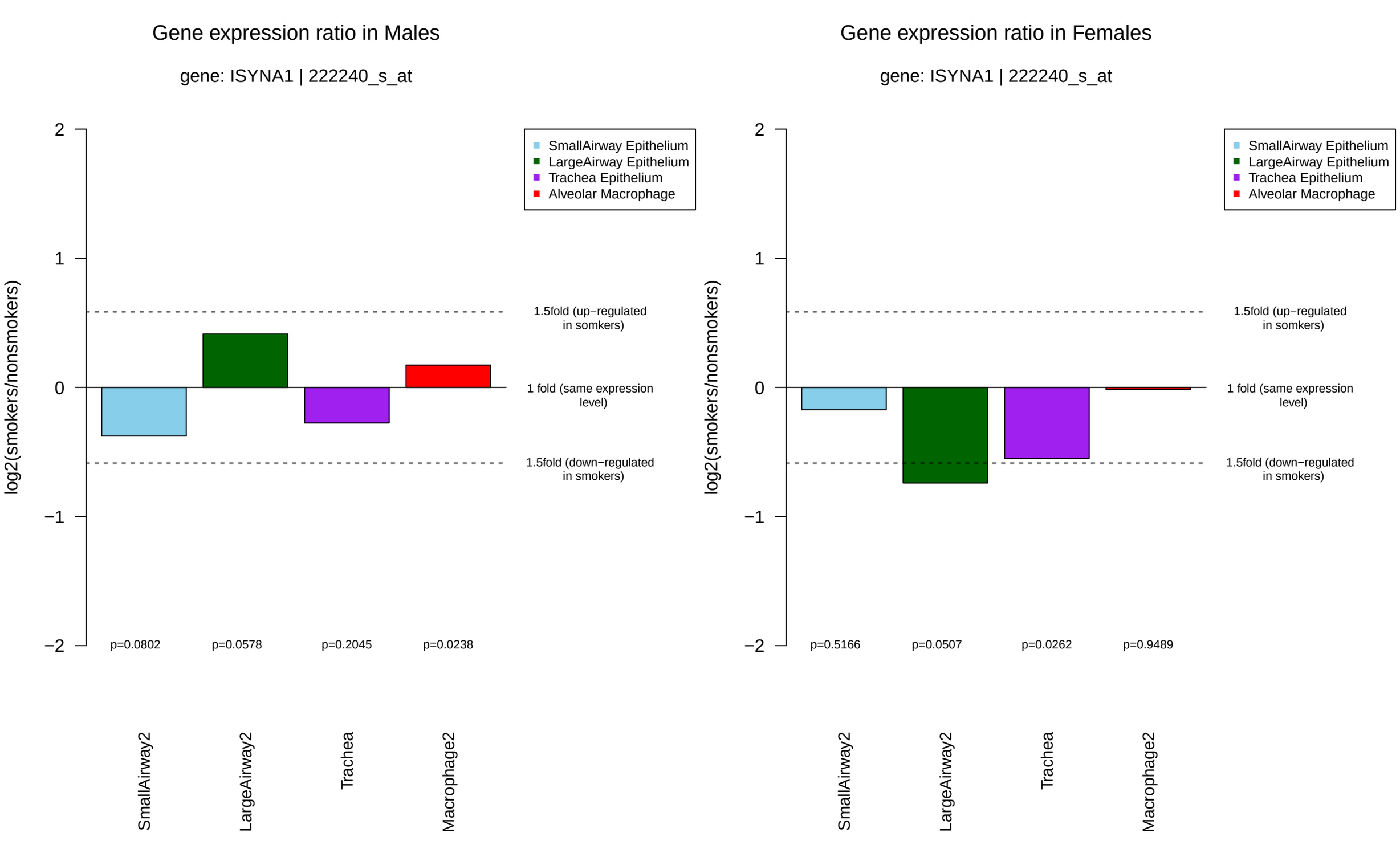
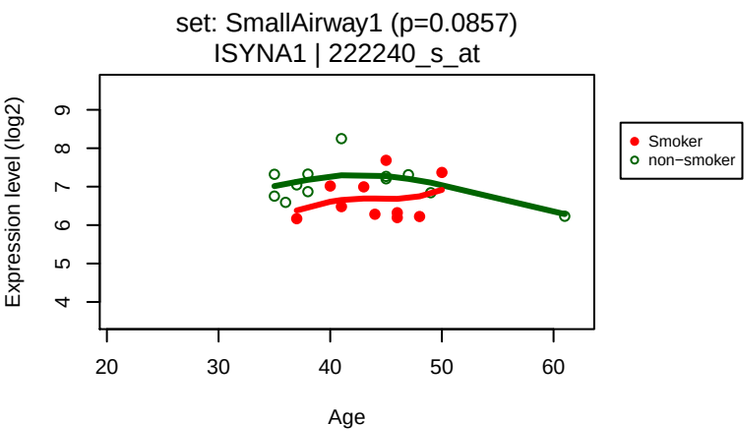
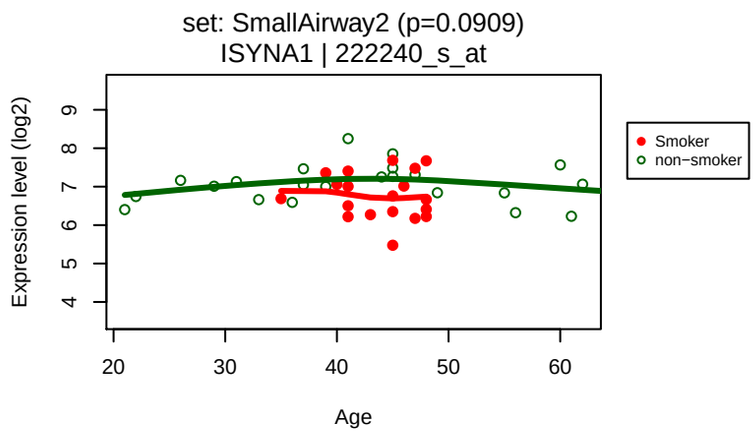
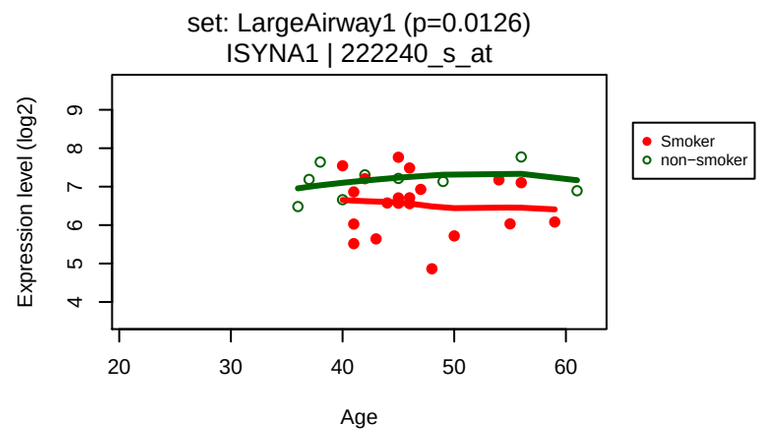

Gene expression of smokers and nonsmokers
gene: ISYNA1 | 222240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0857
p=0.0909
p=0.0224
p=0.0126
p=0.9895
p=0.0257
p=0.524
p=0.2382
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

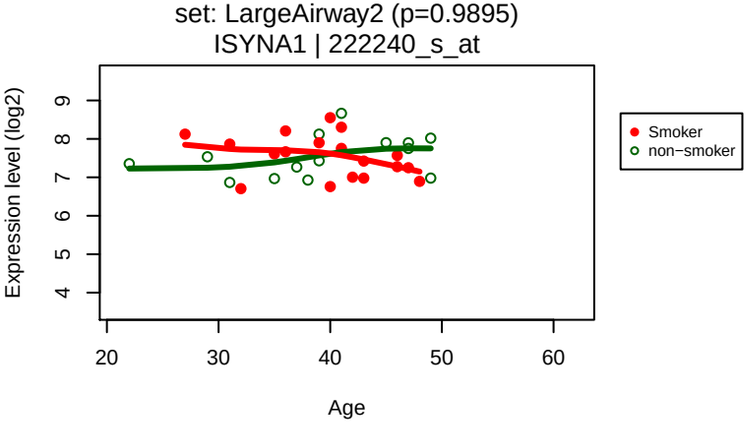

Gene expression ratio (smokers/nonsmokers)
gene: ISYNA1 | 222240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0857
p=0.0909
p=0.0224
p=0.0126
p=0.9895
p=0.0257
p=0.524
p=0.2382
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

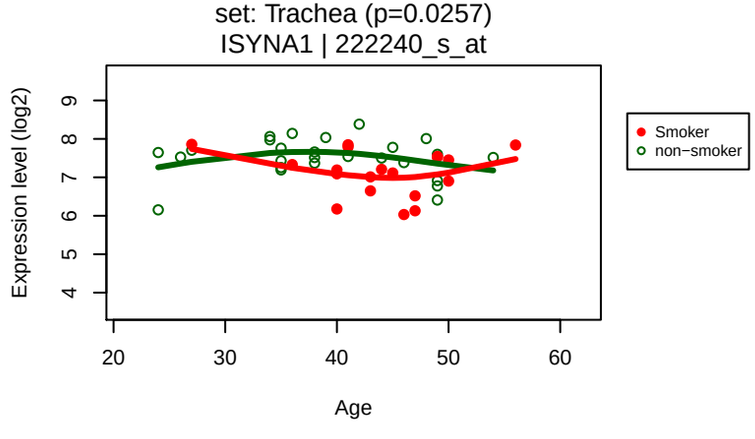

Gene expression ratio in Males
gene: ISYNA1 | 222240_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0802
p=0.0578
p=0.2045
p=0.0238
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISYNA1 | 222240_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5166
p=0.0507
p=0.0262
p=0.9489
SmallAirway2
LargeAirway2
Trachea
Macrophage2

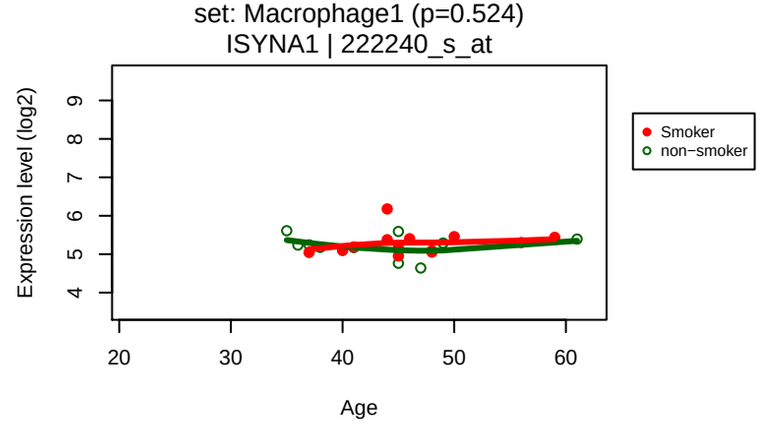

set: SmallAirway1 (p=0.0857)
ISYNA1 | 222240_s_at
Expression level (log2)
Age
Smoker
non−smoker

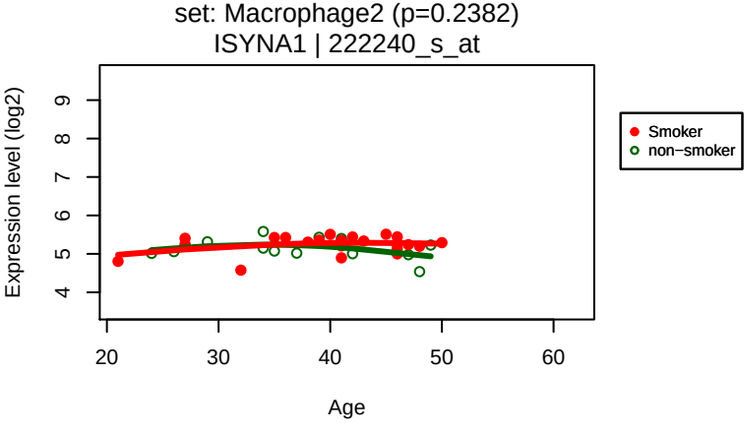

set: SmallAirway2 (p=0.0909)
ISYNA1 | 222240_s_at
Expression level (log2)
Age
Smoker
non−smoker

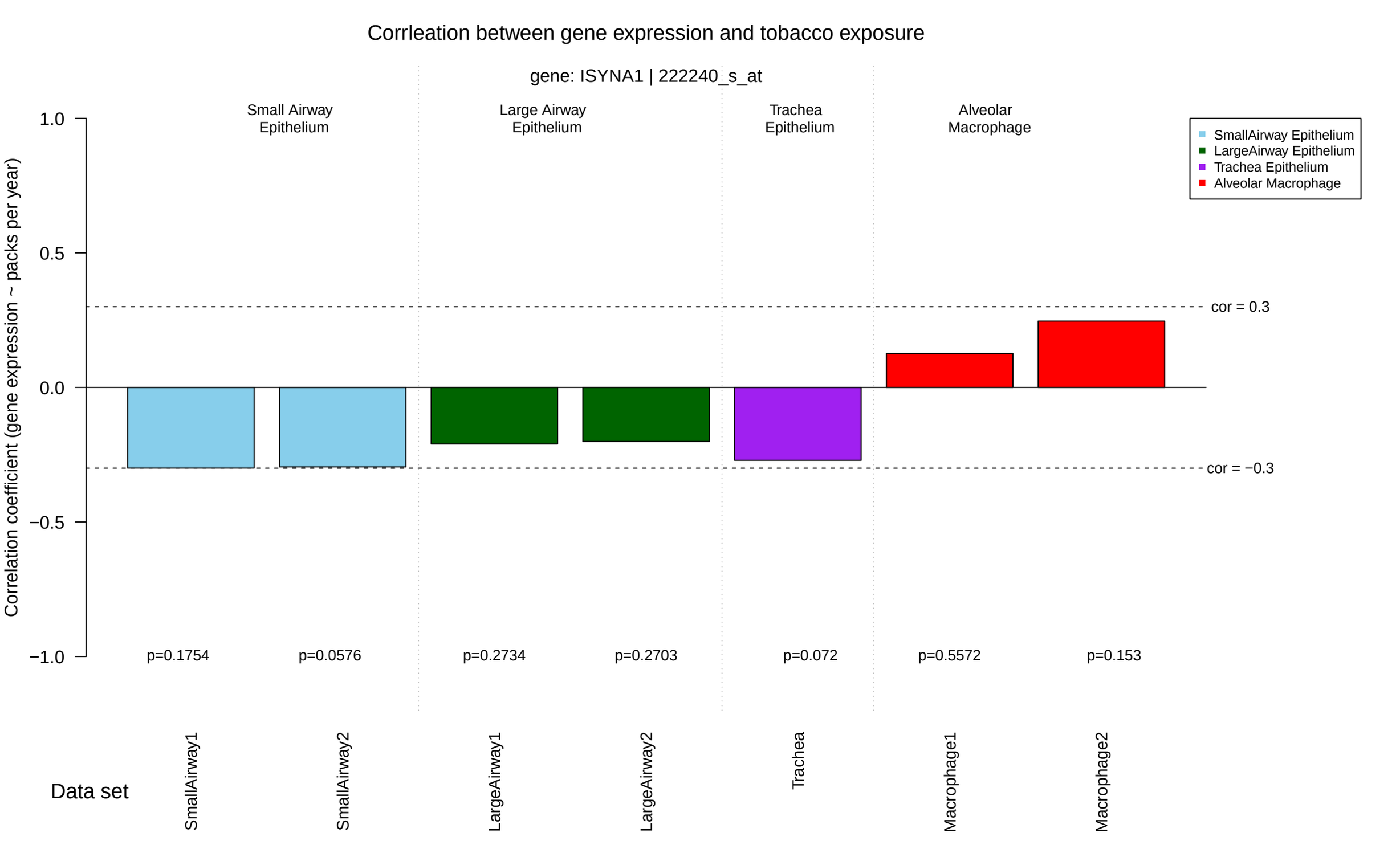

set: LargeAirway1 (p=0.0126)
ISYNA1 | 222240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9895)
ISYNA1 | 222240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0257)
ISYNA1 | 222240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.524)
ISYNA1 | 222240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2382)
ISYNA1 | 222240_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISYNA1 | 222240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1754
p=0.0576
p=0.2734
p=0.2703
p=0.072
p=0.5572
p=0.153
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2